

Atlas-Ingest: Production System Architecture & Design Specification

Document Version: 2.0.0
Target: High-Throughput Distributed Web Ingestion, LLM Extraction & Entity Intelligence
System Type: Multi-Modal Batch & Real-Time Data Pipeline
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1. Executive Summary & System Overview

Atlas-Ingest is an enterprise-grade, multi-modal data ingestion and entity intelligence pipeline designed to continuously harvest, extract, normalize, and resolve structured intelligence from heterogeneous web sources.

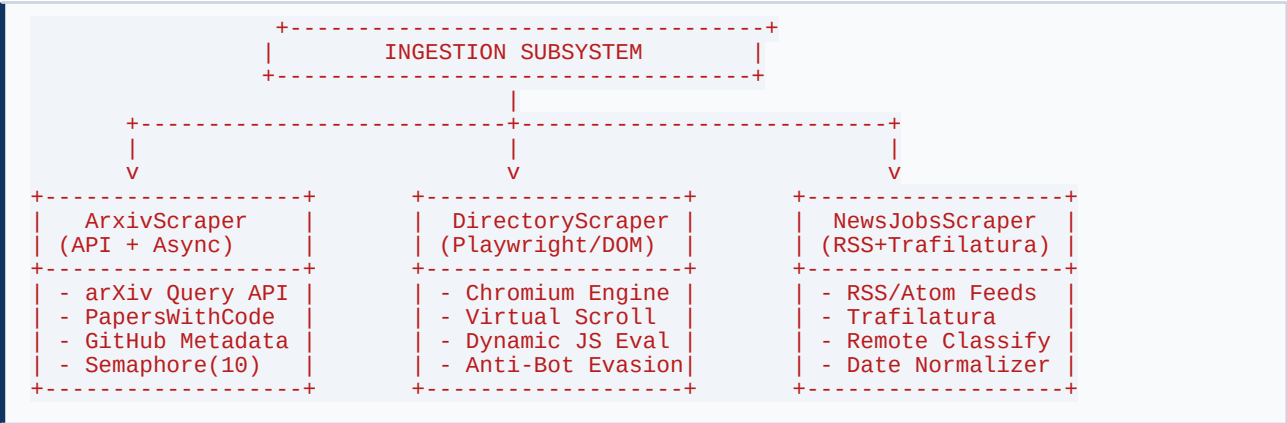
The platform handles both **historical batch extraction** (startups, e-commerce products, arXiv research papers) and **real-time live monitoring** (technology news feeds, global remote tech job boards). Extracted unstructured data is transformed into strictly typed Pydantic models through a resilient multi-tier Large Language Model (LLM) orchestration layer, deduplicated and canonicalized via an asynchronous entity resolution engine, and exported to analytical storage and live Google Sheets dashboards.



2. Detailed Subsystem Architecture

2.1. Ingestion & Crawler Subsystem

The crawler layer comprises specialized, decoupled scraping engines tailored to diverse web transport mechanisms:



ArxivScraper & Academic Correlator ([src/crawlers/arxiv_scraper.py](#)):

- Harvests papers across AI/ML taxonomies ([cs.AI](#), [cs.LG](#), [cs.CV](#), [cs.CL](#), [cs.NE](#)).
- Concurrently enriches abstracts with GitHub repository URLs and code release metadata using PapersWithCode APIs.
- Leverages `asyncio.Semaphore(10)` to bound outbound concurrency, preventing upstream API rate limiting.

DirectoryScraper ([src/crawlers/directory_scraper.py](#)):

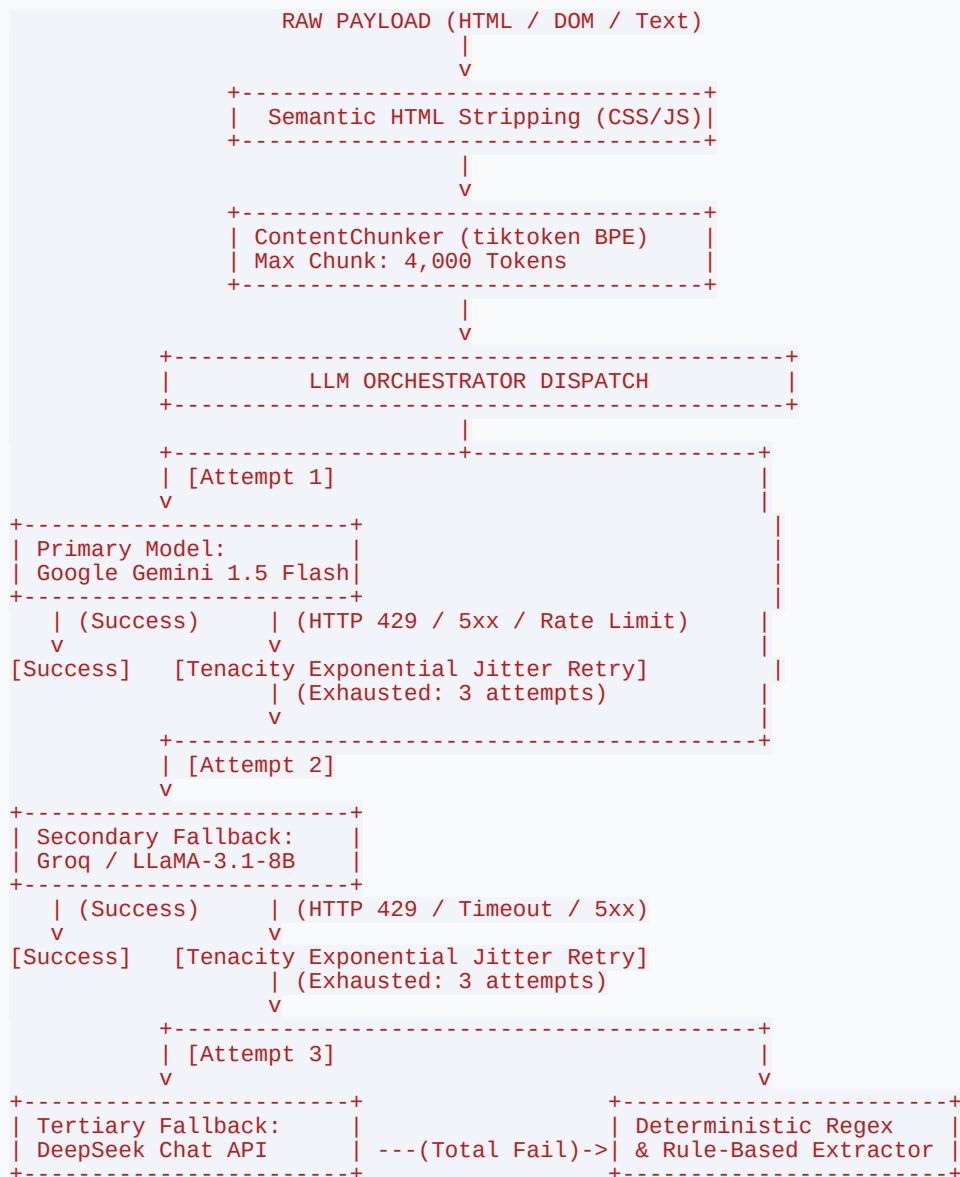
- Utilizes headless Playwright Chromium instances to scrape complex client-side Single Page Applications (SPAs) like Y-Combinator company directories.
- Emulates human interaction through smooth scrolling, viewport resizing, and dynamic network idle timeouts.

News & Jobs Real-Time Scraper ([src/crawlers/news_jobs_scraper.py](#)):

- Polls high-velocity RSS/Atom feeds (TechCrunch, VentureBeat, Hacker News, RemoteOK, WeWorkRemotely, YC Work at a Startup).
- Uses `trafilatura` for noise-free body text extraction, stripping ads, boilerplate, and tracking scripts.
- Applies natural language heuristics to infer remote eligibility (`is_remote: bool`) and salary ranges.

2.2. Resilient LLM Extraction & Fallback Architecture

Unstructured web payloads are converted into structured, typed data through an orchestration layer built on [LiteLLM](#) and [Tenacity](#).

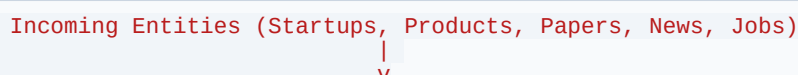


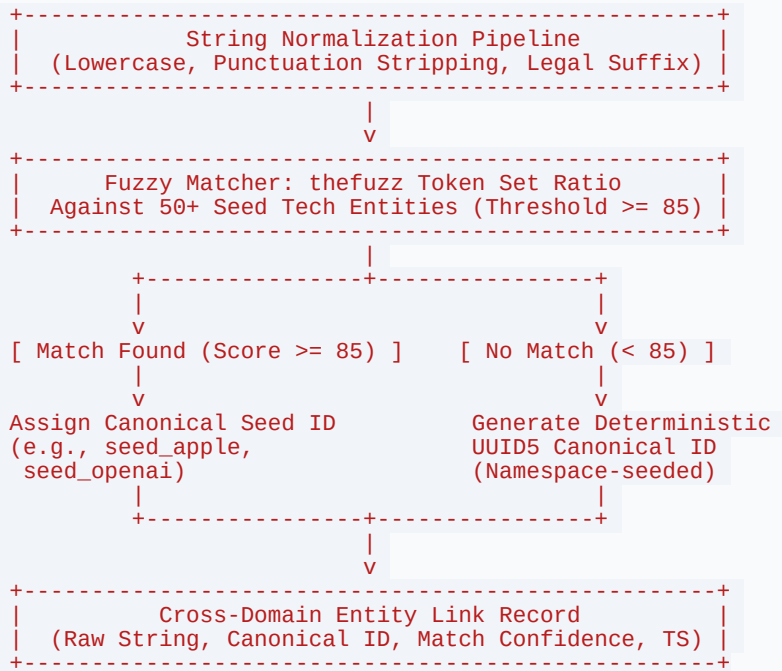
Fault Tolerance Strategies:

- **Context Window Overflow (HTTP 413) Protection:** Raw text is tokenized with `tiktoken`. Strings exceeding LLM context boundaries are split into semantic chunks with 200-token overlaps, extracted independently, and merged.
- **Rate Limit (HTTP 429) & Outage Recovery:** Handled via a multi-tier fallback hierarchy with jittered exponential backoff: `Backoff = random_uniform(0.5 * 2^attempt, 1.5 * 2^attempt)`
- **Strict Typing:** All extractions are parsed directly into Pydantic V2 schemas with automatic type coercion and validation guards.

2.3. Entity Resolution & Cross-Domain Linkage Engine

The **EntityResolver** (`src/resolution/resolver.py`) acts as the linkage bridge, uniting disparate records across all 5 operational verticals into a unified knowledge graph.





- **Seed Knowledge Base:** Contains 50 curated industry giants (OpenAI, Google, Microsoft, Anthropic, Stripe, Shopify, Meta, NVIDIA, etc.).
- **Deduplication:** Prevents duplicate entity representation when company names vary across sources (e.g., "Shopify Inc.", "Shopify Inc", "shopify.com" resolve to canonical `seed_shopify`).

3. Data Model & Schema Specification

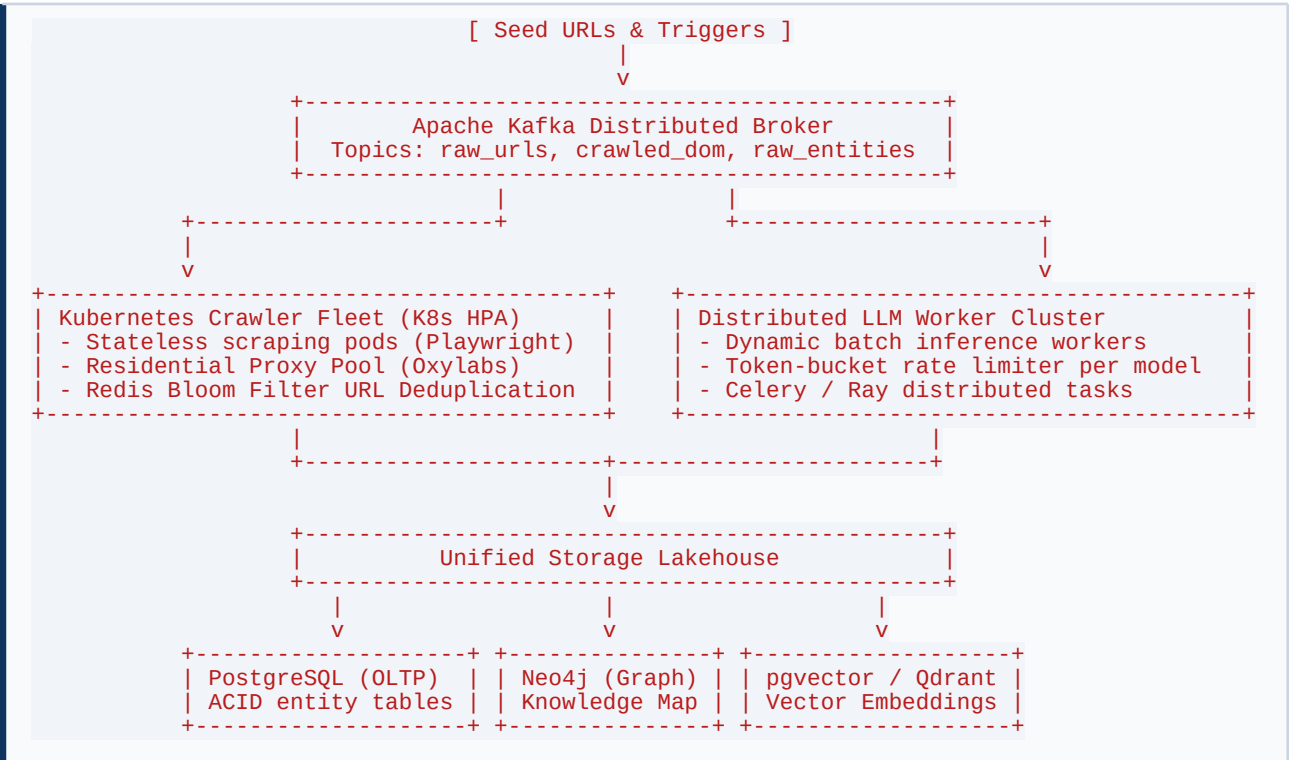
All pipeline artifacts strictly implement validated Pydantic V2 models (`src/models/schemas.py`):

Schema Name	Target Entity	Core Fields	Output Location
StartupEntity	Tech Companies / Startups	name, batch, website, description, founders, tags, canonical_id	output/startups.csv
ProductEntity	E-Commerce Items	title, brand, price, currency, url, category, canonical_id	output/products.csv
ResearchPaperEntity Academic Papers	title, authors, abstract, arxiv_id, published_date, github_repo_url, code_url	output/research_papers.csv	

Schema Name	Target Entity	Core Fields	Output Location
NewsEntity	Tech News Articles	title, source, url, published_at, summary, mentioned_entities, full_text	output/news.csv
JobEntity	Job Postings	title, company, location, is_remote, salary_range, url, source	output/jobs.csv
EntityMapping	Graph Relationships	raw_name, canonical_id, canonical_name, confidence_score, entity_type	output/entity_mappings.csv

4. Production Scale Roadmap (500,000+ Records)

For enterprise-scale production scaling to hundreds of thousands of daily records, Atlas-Ingest transitions to an event-driven, distributed microservices topology:



4.1. Distributed URL Queue & Deduplication

- **Redis Bloom Filters:** Employs an O(1) memory-efficient Bloom Filter holding 10M+ URL signatures with an error probability < 0.01%.

- **Content Hashing:** Generates SHA-256 digests over normalized document text to prevent reprocessing mirrored content.

4.2. Storage Architecture

- **PostgreSQL (JSONB):** Primary relational storage for transactional integrity, polymorphic schema evolution, and fast indexing.
- **Neo4j Graph Database:** Powers multi-hop traversal queries (e.g., *"Find all papers authored by engineers at companies with active job postings for LLM Architects"*).
- **Vector Storage (pgvector):** Generates 1536-dimensional embeddings for semantic similarity search across extracted entities.

5. Technology Stack & Operational Matrix

Component Layer	Technology Selection	Architectural Rationale
Language & Runtime	Python 3.10+ / Asyncio	High-concurrency I/O throughput with native coroutine pipelines
Browser Automation	Playwright (Chromium)	Fast, reliable headless execution with anti-detection handling
HTTP Extraction	HTTPX / Trafilatura	Asynchronous HTTP/2 support + noise-free DOM readability
LLM Gateway	LiteLLM + Tenacity	Unified multi-provider SDK with automatic jittered failover
Entity Resolution	TheFuzz (Levenshtein)	High-accuracy fuzzy string normalization and matching
Data Validation	Pydantic V2 (Rust core)	Ultra-fast schema parsing, coercion, and validation
Export & Sync	gsread + Google Auth	Headless cloud synchronization to Google Sheets
Documentation & CI	xhtml2pdf / GitHub Actions	Automated regression testing and PDF build pipeline